MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 4701: SOFTWARE ENGINEERING

END OF SEMESTER EXAMINATION

DATE: MARCH, 2020 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

1. Answer **ALL FOUR** questions.
2. Marks for each question are indicated.
3. Write your answers clearly.
4. Indicate the question number for each answer.
5. This paper contains **TWO** pages excluding cover page. Please check.

Question One

a. Discuss the following software process models:
   i. Waterfall. **[3 marks]**
   ii. Evolutionary. **[3 marks]**
   iii. Component-base. **[3 marks]**

b. Explain **three** reasons why program inspections are an effective technique for discovering errors in a program. **[6 marks]**

c. What **two** types of error are unlikely to be discovered through inspections? **[4 marks]**

Question Two

a. Explain the differences between verification and validation. **[4 marks]**

b. Discuss **two** strategic options for legacy system evolution. **[6 marks]**

c. Discuss the following types of software maintenance:
   i. Corrective. **[3 marks]**
   ii. Adaptive. **[3 marks]**
   iii. Perfective. **[3 marks]**
   iv. Preventive. **[3 marks]**

d. Discuss risk management process. **[4 marks]**

e. Explain how Software designers can make use of the concept of 'knowledge in the world' in the design of interfaces. **[4 marks]**

Question Three

A company has just designed a user interface that was an utter disaster. You have been brought in as a Software Engineer consultant to explain what they did wrong.

i. Explain **five** design principles of Interface Design. **[15 marks]**

ii. Explain **five** human factors to be considered in Interface Design. **[10 marks]**

Question Four

The bank has many customers who open and maintain one or more accounts. For each Customer, it keeps a record of their name and address. The bank also assign them a unique CustomerID. It assign this unique identifier both for convenience and for accuracy. It is much easier to identify a single customer using their CustomerID rather than by looking up their full name and address. In addition, it is possible for the bank to have two customers with the same name (e.g., Tally Center). In such cases, the unique CustomerID can always be used to tell them apart.

In a similar fashion, all accounts are assigned a unique account number. An account can be either a current account or a savings account. Savings accounts earn interest but the only transactions allowed are deposits, withdrawals and payment of bills (e.g., Electricity). Current accounts do not earn interest. The bank maintain the date that the account was opened. This helps the bank track customers and can be useful for marketing purposes. Finally, it maintain the current balance of an account.

i.	Draw a context diagram.	**[5 marks]**
ii.	Draw level 0 data flow diagram.	**[10 marks]**
iii.	Draw level 1 data flow diagram.	**[15 marks]**

END OF QUESTION PAPER